FORECASTING: PART 2

Seventeenth Lecture by Prof. Uday S. Dixit, Department of Mechanical Engineering, IIT Guwahati



CAUSAL OR EXPLANATORY METHODS

- When enough historical data is present, then based on experience, it is possible to relate forecasts to several factors that cause the trends and fluctuations.
- Thus, if these causal factors are measured and can determine their relationships to the product or service of interest, forecasts of considerable accuracy can be computed.
- Factors used in causal models are of several types:
  - Disposable income, housing starts, inventories, and cost-of-living indices
  - Predictions of dynamic factors and/or disturbances, such as strikes, actions of competitors, and sales promotion campaigns

Types of Causal models

- Regression analysis
- Econometric methods

- Regression analysis
  - Forecasting based on regression methods establishes a forecasting function called a regression equation.
  - The regression equation expresses the series to be forecast in terms of other series that presumably control or cause sales to increase or decrease.

Regression analysis

- Let the dependent variable to be forecasted be *y*.
- The independent variable is *x*. Unlike, the exponential smoothing method, the independent variable is not time.
- Mathematically, the relation is

$$y = a + bx$$

x is the observed value of the independent variable

y is the estimated value of the dependent variable

a is the *y*-axis intercept

b is the slope of the line fitted to the sample data (regression line),

respectively.

- The values of *a* and *b* are such that $\sum e^2$ is minimum, where error *e* is the deviation between predicted and observed value of response.
- These values are given by the following equations:

$$b = \frac{\sum xy - n\bar{x}\,\bar{y}}{\sum x^2 - n\bar{x}^2}$$

$$a = \bar{y} - b\bar{x}$$

Example

- A construction company renovates old homes. The company has found that its dollar volume (total value) of renovation work is dependent on the area payroll. The company revenues and the amount of money earned by wage earners for the last 6 years are provided in the table. Obtain a mathematical relationship to predict sales.

Company's sales (in $ millions), *y*	area payroll (in $ billions), *x*
2.0	2
3.0	3
2.5	4
2.0	2
2.0	1
3.5	7

Solution

Sales, y	Payroll, x	x^2	xy
2.0	2	4	4.0
3.0	3	9	9.0
2.5	4	16	10.0
2.0	2	4	4.0
2.0	1	1	2.0
3.5	7	49	24.5
$\sum y = 15.0$	$\sum x = 19$	$\sum x^2 = 83$	$\sum xy = 53.5$

Solution

$$\bar{x} = \frac{\sum x}{6} = \frac{19}{6} = 3.17$$

$$\bar{y} = \frac{\sum y}{6} = \frac{15}{6} = 2.5$$

$$b = \frac{\sum xy - n\bar{x}\,\bar{y}}{\sum x^2 - n\bar{x}^2} = \frac{53.5 - 6 \times 3.17 \times 2.5}{83 - 6 \times 3.17^2} = 0.26$$

$$a = \bar{y} - b\bar{x} = 2.5 - 0.26 \times 3.17 = 1.68$$

The regression equation is $y = 1.68 + 0.26x$

If it is predicted that area payroll will be \$6 billion next year, sales can be estimated for with the regression equation

Sales = 1.68+0.26×6 = 3.24

Standard Error of the Estimate

- The accuracy of the regression estimates is computed by the standard error of the estimate, $S_{y,x}$
- This computation is called the standard deviation of the regression. It measures the error from the dependent variable, *y*, to the regression line, rather than to the mean.

$$S_{y,x} = \sqrt{\frac{\sum (y - y_c)^2}{n - 2}}$$

y = *y*-value of each data point
y_c = computed (estimated) value from the regression equation
n = number of data points

Standard Error of the Estimate

- For calculation simplicity, an alternative representation of $S_{y,x}$ is

$$S_{y,x} = \sqrt{\frac{\sum y^2 - a\sum y - b\sum xy}{n-2}}$$

Example:

Calculate the standard error of the estimate $S_{y,x}$ of the previous example.

$\sum y^2$ is to be calculated

Solution

Sales, y	Payroll, x	x^2	y^2	xy
2.0	2	4	4	4.0
3.0	3	9	9	9.0
2.5	4	16	6.25	10.0
2.0	2	4	4	4.0
2.0	1	1	4	2.0
3.5	7	49	12.25	24.5
$\sum y = 15.0$	$\sum x = 19$	$\sum x^2 = 83$	$\sum y^2 = 39.5$	$\sum xy = 53.5$

$$S_{y,x} = \sqrt{\frac{\sum y^2 - a\sum y - b\sum xy}{n-2}} = \sqrt{\frac{39.5 - 1.68\times 15 - 0.26\times 53.5}{6-2}} = 0.312$$

The standard error is \$0.312

Coefficients of correlation

- Regression lines are not “cause-and-effect” relationships. They merely describe the relationships among variables.
- The regression equation shows how one variable relates to the value and changes in another variable.
- A way to evaluate the relationship between two variables is to compute the coefficient of correlation. This measure expresses the degree or strength of the linear relationship.
- **Identified as *r***, the coefficient of correlation can be any number between +1 and –1.
- The equation of *r* is

$$r = \frac{n\sum xy - \sum x \sum y}{\sqrt{\left\{n\sum x^2 - \left(\sum x\right)^2\right\}\left\{n\sum y^2 - \left(\sum y\right)^2\right\}}}$$

Coefficient of correlation

- Although the coefficient of correlation is the measure most commonly used to describe the relationship between two variables, another measure does exist.
- It is called the coefficient of determination and is simply the square of the **coefficient of correlation r^2.**
- **The value of r^2 will always be a positive number in the range $0 \leq r^2 \leq 1$.**

Multiple-Regression Analysis

- Multiple regression is an extension of the simple regression model. It allows building a model with several independent variables instead of just one variable.

Multiple Linear Regression Analysis

- If a dependent variable y is the function of m independent variables, *viz.*, $x_1, x_2, \ldots\ldots, x_m$, then the multiple linear regression model may be written as

$$y = c_0 + c_1x_1 + c_2x_2 + c_3x_3 + \ldots\ldots\ldots\ldots\ldots\ldots + c_mx_m + e$$

where e is the error term. In the case where, there are n observations, the following n equations can be written:

$$y_1 = c_0 + c_1x_{11} + c_2x_{12} + c_3x_3 + \ldots\ldots\ldots\ldots\ldots\ldots + c_mx_{1m} + e_1,$$
$$y_2 = c_0 + c_1x_{21} + c_2x_{22} + c_3x_{23} + \ldots\ldots\ldots\ldots\ldots\ldots + c_mx_{2m} + e_2,$$
$$\vdots$$
$$y_n = c_0 + c_1x_{n1} + c_2x_{n2} + c_3x_{n3} + \ldots\ldots\ldots\ldots\ldots\ldots + c_mx_{nm} + e_n,$$

where y_i indicates the i-th observed value of y and x_{ij} indicates i-th observed value of x_j.

Multiple Linear Regression Analysis

- In the matrix form, the set of equations can be written as

$$\begin{Bmatrix} y_1 \\ y_2 \\ . \\ . \\ y_n \end{Bmatrix} = \begin{bmatrix} 1 & x_{11} & x_{12} & . & . & x_{1m} \\ 1 & x_{21} & x_{22} & . & . & x_{2m} \\ . & . & . & . & . & . \\ . & . & . & . & . & . \\ 1 & x_{n1} & x_{n2} & . & . & x_{nm} \end{bmatrix} \begin{Bmatrix} c_0 \\ c_1 \\ . \\ . \\ c_m \end{Bmatrix} + \begin{Bmatrix} e_1 \\ e_2 \\ . \\ . \\ e_m \end{Bmatrix},$$

or

$$\mathbf{Y = XC + E}$$

Multiple Linear Regression Analysis

- Our attempt should be to minimize the error vector **E.** If all the terms in the vector **E** are zero, the model is perfect. Usually we minimize **E** in least square sense.

- The sum squared error is given by

$$\mathbf{E^TE = (Y-XC)^T\,(Y-XC) = Y^TY-Y^TXC-C^TX^TY+C^TX^TXC}$$

- By making use of the property that the transpose of a scalar is equal to the scalar itself, we can write

$$\mathbf{Y^TXC = C^TX^TY}\ .$$

- Therefore,

$$\mathbf{E^TE = Y^TY-2C^TX^TY + C^TX^TXC}$$

Multiple Linear Regression Analysis

- Minimizing the last expression respect to **C**, we get

$$\frac{\partial(\mathbf{E}^{\mathbf{T}})}{\partial \mathbf{C}} = -2\,\mathbf{X}^{\mathbf{T}}\mathbf{Y} + 2\mathbf{X}^{\mathbf{T}}\mathbf{X}\mathbf{C} = 0$$

The error will be minimized, if

$$\mathbf{X}^{\mathbf{T}}\mathbf{X}\mathbf{C} = \mathbf{X}^{\mathbf{T}}\mathbf{Y} \qquad (1)$$

or

$$\mathbf{C} = (\mathbf{X}^{\mathbf{T}}\mathbf{X})^{-1}\,\mathbf{X}^{\mathbf{T}}\,\mathbf{Y} \qquad (2)$$

- The coefficient vector can be found either by solving for **C**, from Equation 1 using any equation solver routine or from Equation 2.

- The matrix $(\mathbf{X}^{\mathbf{T}}\mathbf{X})^{-1}\mathbf{X}^{\mathbf{T}}$ is called the pseudo-inverse of **X**.

Non-Linear Regression Analysis

- The procedure for multiple linear regression can also be applied for non-linear regression, when *y* can be expressed as a polynomial function of the dependent variables.

Non-Linear Regression Analysis

- It should be noted that the fitted multiple regression model should also be tested for some data which have not participated in the fitting of the model. For knowing the performance of the fitting, coefficient of determination (R^2) can be calculated given by

$$R^2 = \frac{\sum_{i=1}^{n}(y_i - \bar{y})^2 - \sum_{i=1}^{n}(y_i - \hat{y})^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2}$$

where $\bar{y}$ is the mean and $\hat{y}$ is the predicted value.

R^2 lies between 0 and 1. The closer the value to 1, the better the fitting.

Fuzzy Regression Analysis

- A regression model can also be developed for predicting the lower and upper bounds of the dependent variable.
- With this, *y* can be represented as a triangular fuzzy number by assigning a membership grade of 0.5 to the lower and upper bound estimates.
- A membership grade of 1 to the most likely estimate obtained by a standard multiple regression procedure.

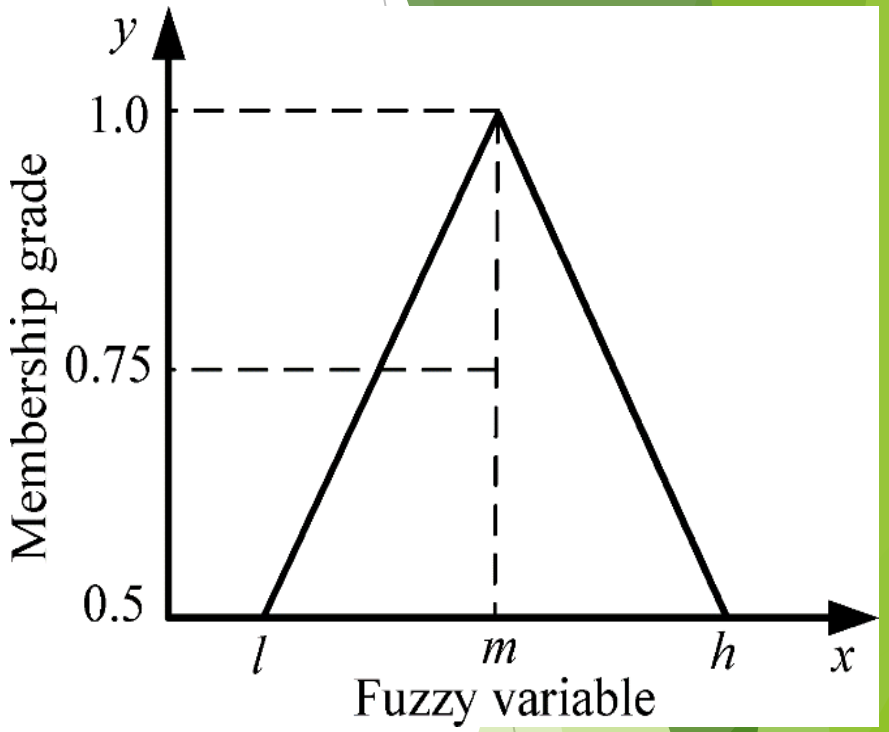


Figure A triangular membership function

- **Fuzzy number**: Generalisation of an interval number in which interval varies with the membership grade.

Non-Linear Regression Analysis

- The lower and upper estimates of the variable can be expressed as

$$y_l(x) = c_0^l + c_1^l x_1 + \ldots\ldots\ldots + c_m^l x_m$$

$$y_u(x) = c_0^u + c_1^u x_1 + \ldots\ldots\ldots + c_m^u x_m$$

where $c_0^l,\ c_1^l, c_2^l \ldots\ldots\ldots, c_m^l$ are the coefficients for predicting the lower estimate and $c_0^u,\ c_1^u, c_2^u \ldots\ldots\ldots, c_m^u$ are the coefficients for predicting the upper estimate of *y*.

Non-Linear Regression Analysis

- Assume that *n* observations are available. The coefficients can be obtained by solving the following optimization problem:

$$\text{Minimize} \sum_{i=1}^{n} \left(c_0^u - c_0^l \right) + \left(c_1^u - c_1^l \right) x_{i1} + \ldots\ldots + \left(c_m^u - c_m^l \right) x_{im},$$

subject to

$$\left.\begin{aligned} c_0^l + c_1^l x_{i1} + \ldots\ldots\ldots + c_m^l x_{im} \le y_i \\ c_0^u + c_1^u x_{i1} + \ldots\ldots\ldots + c_m^u x_{im} \ge y_i \end{aligned}\right\} \quad \text{for } i = 1, 2, \ldots., n.$$

This ensures that the sum of the differences between the upper and lower estimates is minimized.

QUALITATIVE OR JUDGMENTAL METHODS

- Sometimes, the past data, even when they exist, may not be representative of future conditions.
- Estimates of what people think, samplings of how they react to market tests, knowledge of consumer behavior, and analogies to similar situations may be the best that management can do.

QUALITATIVE OR JUDGMENTAL METHODS

- The qualitative methods are of considerable significance, then, because they provide a basis for some important decisions.

  - The Delphi Method
  - Market Surveys
  - Historical Analogy and Life Cycle Analysis
  - Scenario Based Forecasting
  - Sales force composite

The Delphi Method

- There are three different types of participants in the Delphi method: **decision makers, staff personnel, and respondents**.
- Decision makers usually consist of a group of 5 to 10 experts who will be making the actual forecast.
- The panel of experts can be organized in various ways, and it often includes individuals from both inside and outside the organization.
- Each panel member is an expert on some aspect of the problem, but no one is an expert on the entire problem.

The Delphi Method

- The procedure in Delphi Method involves the following:

1. Each expert in the group makes independent predictions in the form of brief statements.

2. The coordinator edits and clarifies these statements.

3. The coordinator provides a series of written questions to the experts that include feedback supplied by the other experts.

4. Steps 1 to 3 are repeated several times. In practice, convergence is usually obtained after a small number of rounds.

Market Surveys

- This method solicits input from customers or potential customers regarding future purchasing plans.
- It can help not only in preparing a forecast but also in improving product design and planning for new products.
- The consumer market survey and sales force composite methods may suffer from overly optimistic forecasts that arise from customer input.
- This method involves the use of questionnaires, consumer panels, and tests of new products and services

Historical Analogy and Life Cycle Analysis

- Market research studies can sometimes be supplemented by referring to the performance of an ancestor of the product or service under consideration and applying an analysis of the product life cycle curve.
- For example, the assumption can be made that smart mobile would follow the general sales pattern experienced with keypad mobile. However, it would take twice as long to reach a steady state.
- Such comparisons provide guidelines during initial planning phases and may be supplemented by other kinds of analyses and studies as actual demand becomes known.

Scenario Based Forecasting

- A fundamentally different approach to judgmental forecasting is scenario-based forecasting. The aim of this approach is to generate forecasts based on reasonable scenarios.
- The scenarios are generated by considering all possible factors, their relative impacts, the interactions between them, and the targets to be forecast.
- Building forecasts based on scenarios allows a wide range of possible forecasts to be generated and some extremes to be identified.
- For example it is usual for “best”, “middle” and “worst” case scenarios to be presented.
- Thinking about and documenting these contrasting extremes can lead to early contingency planning.

Sales force composite

- In this approach, each salesperson estimates what sales will be in his or her region.
- These forecasts are then reviewed to ensure that they are realistic. Then they are combined at the district and national levels to reach an overall forecast.
- For example, a variation of this approach occurs at Maruti/Hyundai, where every quarter these dealers have a “make meeting.”
- At this meeting, they talk about what is selling, in what colors, and with what options. This way the factory knows what to build in the near future.